



Use of causal inference methods in case-control studies: a methodology review

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Abstract

Background: The use of causal inference methods in cohort studies has increased considerably in recent years. However, their use has been limited in case-control studies. This report aimed at providing a detailed review of causal inference methods used in case-control studies and to review and examine their applications in previous studies.

Methods: Several methods have been used to facilitate causal inference in case-control studies, including intercept-adjustment, propensity scores, weight-based and doubly-robust estimators. We used the Medical Literature Analysis and Retrieval System Online database to identify original peer-reviewed case-control studies conducted from March 2014 to March 2024 that applied these methods.

Results: We identified 418 studies, 23 of which met the inclusion criteria. Most studies involved case-control matching (individual or frequency) and included incident cases. The covariate-conditional odds ratio was the most frequently reported estimated parameter. Sixty-five percent of included studies considered an adjustment for sampling bias, most often using inverse-probability of observation weighting and case-control targeted maximum likelihood approaches.

Conclusion: We are still in early stages of development and application of causal inference methods for case-control studies. Their implementation and new techniques to address time-varying confounding can improve the validity of study findings and should be encouraged.

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INTRODUCTION

Causal inference methods are a loose collection of statistical approaches that have been shown, under varying sets of assumptions, to directly estimate causal effects. In the potential outcomes framework, a causal effect can be expressed as the contrast in outcomes that would be expected between two (hypothetical) copies of the same population, that are identical except that they differ in the exposure of interest. Randomized controlled trials (RCTs) are the gold standard method for investigating many causal questions, as weaker assumptions are required for causal interpretation of a given estimate due to the study design and execution. However, for many scenarios, RCTs are not ethical or lack generalizability to target populations of interest,¹ which highlights the importance of methods to facilitate causal inference from observational studies. Although the last few decades have seen a meteoric rise in the development and application of causal inference methods for observational studies, the focus has primarily been on cohort studies. However, cohort studies can be impractical for studying rare outcomes or when exposure assays are costly.² In these scenarios, case-control study designs are often used for cost-efficiency.

By far, the most commonly used method for estimating exposure effects in case-control studies is logistic regression, in which case the case-control odds ratio (OR) is often the *de facto* estimand of interest.^{2,3} The case-control OR is commonly used because, under cumulative sampling that is nested in a cohort, the case-control OR estimates the OR that would be obtained in a cohort study.⁴ Thus, in some situations, case-control studies and cohorts give consistent inference on ORs, and if causal assumptions are met, such ORs can represent causal ORs.⁵ However, logistic regression parameters are interpretable as (at best) conditional causal effects, given a measured set of adjustment variables. Causal methods are useful for estimating marginal causal parameters which correspond to population average effects like the marginal risk differences, which equal the average of individual level causal effects in the entire population of interest and have a strong implication for public health.⁶ Thus, causal inference methods can be useful both for addressing bias as well as estimating parameters for specific populations, ideas that have previously been demarcated as internal and external validity.

Several causal inference methods for case-control studies have been introduced,^{7–9} but are rarely used in practice.¹⁰ To facilitate broader adoption and clarify their purpose, we review available methods proposed for causal inference in case-control studies and their associated assumptions. We discuss applications in the literature, provide code, and weigh the strengths and weakness of each method to identify gaps and provide recommendations. Brief introductions of causal inference and design of case-control studies is presented in the Supplementary material (Appendix S1).

METHODS

Since Rubin's framing of causal inference in observational studies through potential outcomes in 1974,¹¹ many methods have been developed under this framework for cohort studies. Commonly used methods included propensity scores/ inverse probability of treatment weighting (IPTW), g-computation and doubly-robust methods (such as targeted

maximum likelihood estimation). Each of these methods focuses on statistical control of measured confounders and ensuring results are relevant to a specific target population. For conciseness, we do not consider other approaches such as effect decomposition (causal mediation analysis)^{12,13}, negative controls¹⁴ or instrumental variable methods¹⁵ like Mendelian randomization that use alternative assumption sets.

For case-control studies, the majority of these methods have been adapted to control confounding (internal validity) while also addressing the issue of sampling bias (external validity). By “sampling bias,” we mean bias that occurs because non-cases are under-sampled from the target population, relative to cases. Sampling bias manifests in two ways: 1) prevalence in the study population is high relative to the target population (“outcome sampling bias”) and 2) covariates that are associated with the outcome will have different distributions in the study population relative to the target population (“covariate sampling bias”). Here, we review methods that typically fall under the umbrella term “causal inference approaches” or “g-methods”, including propensity scores,^{16,17} case-control weighted estimators (CCW-MLE),^{8,18} and case-control targeted maximum likelihood estimators (CCW-TMLE)^{8,18–20}. In addition, we review the more classical method of intercept-adjusted logistic regression,^{21,22} which can also be used to address sampling bias while addressing confounding, such that this approach falls under the “outcome regression” or “g-computation” family of causal inference approaches. More technical illustration of sampling bias can be found elsewhere.^{7,8,10}

The causal-inference methods we review here have been developed primarily for cumulative incidence sampling, where cases and controls (non-cases) are selected at the end of follow-up. Some approaches also work with case-cohort designs where non-cases are replaced with a sample (at baseline) from a cohort within which the cases are nested²³. Incidence density sampling, in which controls are selected from among those at risk at the time each case arises, is briefly considered in the discussion.

In the following, we consider case-control data in which we have sampled cases and controls in the target population (Pop). For simplicity, we assume that the distribution of the observed outcome Y can be described using a logistic regression model:

$$\log\left(\frac{P(Y=1|\mathbf{X}, Pop)}{P(Y=0|\mathbf{X}, Pop)}\right) = \beta_0 + \sum_{j=1}^p \beta_j X_j$$

where $\mathbf{X} = (X_1, \dots, X_p)$ is a vector of p covariates (which includes the exposure A) sufficient to control confounding; β_0 and β_j for $j = 1, \dots, p$ are the log-OR parameters to be estimated.

Intercept-adjusted logistic regression

A simple approach to address outcome sampling bias has long been known for unmatched case-control studies,^{22,24,25} which allows the estimation of absolute risk. It involves adding an offset term c_0 to the standard logistic regression.^{22,24,25} This is equal to the logarithm of the sampling OR:

$$\log(c_0) = \log \left(\frac{\frac{P(Y=1|Pop)}{P(Y=0|Pop)}}{\frac{P(Y=1|Study)}{P(Y=0|Study)}} \right)$$

where $P(Y=1|Pop)$ and $P(Y=1|Study)$ represent the prevalence of the outcome in the target population and the case-control study data, respectively.

This approach, which requires knowledge of the population prevalence of the outcome, makes it possible to estimate some population level causal effects.^{18,26} Matching can also be accommodated using stratum-specific sampling ORs.²⁵ Given sufficient control of confounders, the intercept-adjusted logistic model will estimate the conditional causal OR (an OR for a group of individuals with a specific set of covariates). It can be used to estimate marginal causal effects on other scales (e.g. risk difference), but not for the target population due to residual covariate sampling bias.

To estimate marginal effects in the target population, intercept adjustment must also be accompanied by weighting or standardization to the covariate distribution of the target population. This holds because effect measures vary over the covariate distribution in the study, which is subject to sampling bias except when the covariates are not associated with the outcome. This issue also arises when oversampling certain groups in case-control studies, for example if racial or ethnic minorities are over-sampled to power the estimation of disparities. In matched studies, one must weight or standardize across the distribution of matching variables.^{18,26} Case-control sample weighting can address this issue, as we discuss below.

Propensity scores

While logistic regression addresses confounding through regression adjustment, propensity score method, a widely used set of methods, rely on using a propensity score (a covariate-conditional probability of exposure) for addressing confounding.¹⁷ To control confounding, different propensity score methods can be used: propensity score matching, IPTW using the propensity score, stratification on the propensity score and covariates adjustment using the propensity score.^{16,17} These methods vary with respect to whether and how sampling bias is addressed.

Briefly, the propensity score is the probability of being exposed, given other covariates in the population:

$$\pi = P(A=1|Z, Pop)$$

Where Z is the covariate vector X excluding exposure A . For estimating average effects of binary exposures, an inverse probability weight can be defined for each individual as follows:

$$w_i = \begin{cases} \frac{1}{P(A_i = 1 | Z_i, Pop)}, & \text{if the individual is exposed} \\ \frac{1}{1 - P(A_i = 1 | Z_i, Pop)}, & \text{if the individual is unexposed} \end{cases}$$

Thorough introductions to propensity scores can be found elsewhere.^{16,27} Crucially, for case-control studies, the sample estimate of the propensity score $P(A = 1 | Z, Study)$ is subject to sampling bias because, for example, if the exposure increases the probability of the outcome, given Z , exposed individuals will be more common in the case-control study relative to the cohort study. Generally, propensity score methods (and other causal inference methods) can rely on bootstrapping to generate standard errors, but specific estimators may allow less computationally intensive approaches.

Multiple ways of addressing sampling bias in the propensity score have been proposed for case-control studies.^{28–31} Under rare outcomes, Robins proposed that the IPTW can be used in the context of marginal structural models to estimate marginal ORs in the target population²⁸ with the propensity score estimated only among controls, thus addressing covariate sampling bias but not outcome sampling bias. The weight is then applied to all individuals and allows that the population risk of the outcome can be unknown (aside from knowing that it is rare).²⁸ Mansson introduced several novel approaches to use of propensity scores in case-control studies, which involved a) estimating propensity scores in a subset of the study that included all controls and a subset of cases and b) including case status in the propensity score model and estimating propensity scores for all individuals as though they had been controls.²⁹ Similar to Robins' approach, both approaches are restricted to the estimation of the marginal OR because they address covariate sampling bias only through the propensity score estimates. The approach proposed by Zhu et al. estimates the propensity score model among cases and then matches controls on the predicted propensity scores before estimating conditional log-ORs.³¹ However, this approach does not yield valid estimates of any marginal parameter, so it is only tangentially related to causal inference vis-a-vis adjustment for confounding or sampling bias.

G-computation

G-computation, as a complement to propensity score methods which model exposures, consists of the modelling outcomes and predicting potential outcomes under different exposure conditions.^{32,33} For time-fixed exposures, the main steps of g-computation are:

1. Regress the exposure and the covariates on the outcome;
2. Using the regression model, predict outcomes for individuals under each exposure condition;
3. Estimate average causal effects by averaging the predicted outcomes across the study population.

The third step is equivalent to standardizing over the covariate distribution of the study population, which yields study-population averaged effects. Confounding and sampling bias are addressed in steps 1 and 3 (described below). If the study population is a representative

sample of the target population, then this estimates a population average (marginal) causal effect. For time-varying exposures with time-varying confounders impacted by prior exposures, the algorithm is more complicated (e.g.^{34,35}).

The case-control weighted maximum likelihood estimator (CCW-MLE) is useful as a step in g-computation. CCW-MLE uses weighting similar to one of the methods demonstrated by Mansson et al.²⁹ This approach aligns with Miettinen's standardization approach³⁶ as applied by Newman.³⁰

A weighted logistic regression is preferably used in the first step with $P(Y = 1)$ and $\frac{P(Y = 0)}{J}$ (which are assumed known from the target population of the case control study participants and J is the ratio of controls to cases) respectively the weights for cases and controls for unmatched studies.^{10,20} Intuitively, the weighted proportion of cases (and weighted distribution of covariates) in the case-control data is then equal to the marginal proportion of cases (and marginal distribution of covariates) in the target population, so that more general models (outside logistic regression) can be fitted to the data and marginal effects can be estimated. Then, g-computation steps 2 and 3 from above can then be followed, with the modification that step 3 uses case-control weighted averages to fully account for covariate sampling bias. Bootstrap¹⁹ and delta²⁶ methods can be used to obtain standard errors. Matching can also be accommodated.²⁰ This method allows the estimation of marginal effects (e.g. ORs, relative risks, and risk differences).

Targeted maximum likelihood estimation (TMLE)

TMLE is a causal inference estimation method that can use models for the exposure and the outcome. TMLE is doubly-robust, meaning it has the advantage of providing a consistent estimator if either model is consistent, and can have improved variance if both models are consistent.³⁷ The typical steps involve getting a first stage estimate of the outcome risk (like g-computation) and then updating that estimate using a modification of the propensity score. TMLE and other doubly-robust approaches are discussed thoroughly elsewhere.^{38,39}

Specifically, the steps to estimate the case-control weighted TMLE estimator are as follows:⁸

1. Define case-control weights for cases and controls as in CCW-MLE. In the following, we will define $q_0 \equiv P(Y = 1)$ (such that $1 - q_0 = P(Y = 0)$).
2. Fit a CCW-MLE or intercept-adjusted logistic regression model predicting the outcome based on the exposure and covariates and calculate the initial probabilities of the outcome for each individual under each exposure condition, denoted by $\widehat{Q}(A, Z)$.
3. Fit a CCW-MLE logistic regression model for A to estimate propensity scores, denoted by $\widehat{g}(A, Z)$.
4. Using the propensity scores, compute "clever covariates" for each individual. A clever covariate is a covariate that is used to adjust or update the prediction $\widehat{Q}(A, Z)$ based on the estimated propensity scores. This allows the incorporation

of information about how covariates relate to the exposure. For a risk difference, the clever covariate is univariate and is given by:

$$h(A, Z) \equiv \left(\frac{I(A=1)}{\pi(A=1|Z)} - \frac{I(A=0)}{\pi(A=0|Z)} \right),$$

where $I(A=a)$ is an indicator function that equals 1 when $A=a$ and zero otherwise. For relative risk and OR, the bivariate clever covariate vector is given as the set of disjoint indicators:

$$h(A, Z) \equiv \left(\frac{I(A=1)}{\pi(A=1|Z)}, \frac{I(A=0)}{\pi(A=0|Z)} \right).$$

5. Update the probabilities $\widehat{Q}(A, Z)$ through a CCW-MLE logistic regression model with $h(A, Z)$ as a supplementary variable to estimate a fluctuation parameter ε by regressing the observed outcome on the clever covariate(s) in an intercept-free logistic model with the initial prediction $\widehat{Q}(A, Z)$ used as an offset term. The initial predictions are then updated:

$$\widehat{Q}_1(A, Z) = \ell^{-1} \left(\ell \left(\widehat{Q}(A, Z) \right) + h(A, Z) \hat{\varepsilon} \right)$$

Where ℓ and ℓ^{-1} are the logistic link function and its inverse.

6. Compute the targeted estimate of interest using the updated probabilities by averaging over the CCW-MLE distribution of the covariates. For example, the marginal causal risk ratio can be computed as follows:

$$\widehat{\psi}_{RR} = \frac{\frac{1}{n} \sum_{i=1}^n \left[I(Y_i=1) q_0 \widehat{Q}_1(1, Z_i) + I(Y_i=0) \frac{1-q_0}{J} \widehat{Q}_1(1, Z_i) \right]}{\frac{1}{n} \sum_{i=1}^n \left[I(Y_i=1) q_0 \widehat{Q}_1(0, Z_i) + I(Y_i=0) \frac{1-q_0}{J} \widehat{Q}_1(0, Z_i) \right]}$$

Where $I(Y_i=y)$ is an indicator function that equals 1 if $Y_i=y$, and 0 otherwise and the summation is over the n participants comprising cases and controls.

Standard errors can be calculated using influence curve¹⁹ or bootstrap-based methods. Modifications to TMLE can enhance power and stability.^{9,10} An alternative to TMLE and other doubly-robust methods often employed in cohort studies is to use propensity score and outcome regression based methods to estimate the same (or similar) quantities, estimates of which can then be contrasted to assess potential impacts of model specification. Case-control studies can (and have) employed similar strategies, but must take care that sampling bias is handled similarly in each approach.

REVIEW OF PUBLICATIONS USING CAUSAL INFERENCE METHODS WITH A CASE-CONTROL DESIGN

Published studies of English language were identified using Medical Literature Analysis and Retrieval System Online (MEDLINE) database. We included original case-control studies published from March 12, 2014 to March 11, 2024. Mendelian randomization, genome-wide association studies and mediation analyses were excluded. The search strategy used is presented in the Appendix S2, but the goal was to identify case-control studies that explicitly mentioned causal inference.

We used Covidence (Veritas Health Innovation, Melbourne, Australia), an online software for the management of systematic reviews to screen and review the studies. Two authors (MM and MX) independently screened and reviewed each abstract and full text. Disagreements were solved by discussion to reach consensus. We then extracted data from all included studies according to: i) type of case-control study, ii) recruitment of cases, iii) software used; iv) type of paper, v) parameter estimated, vi) methods used, vii) discussion of causal assumptions and viii) adjustment for sampling bias.

Our search yielded 418 studies, 47 of which were assessed for full-text eligibility (Figure 1). Of the 23 included studies, five were from the United States, five (including one re-analysis) from Iran and five from Europe. Table 1 presents a summary of the included studies. More than half of the studies were published from 2020 and onwards. Twenty-six percent of the studies were nested case-control studies, 65% were matched and 74% included incident cases. Most studies (83%) were applied, as opposed to methodological studies (17%). The OR was the most frequently reported estimated parameter (74%). Over 50% of studies performed propensity methods/IPTW, while approximately 30% used CCW-TMLE. At least one causal assumption (e.g. exchangeability, causal consistency, positivity, or variations of those terms) was explicitly discussed in only 56% of studies, and 65% mentioned adjustment for sampling bias from the case-control design.

DISCUSSION

In this paper, we reviewed existing methods that have been proposed for explicitly estimating marginal causal effects in case-control studies in which cases and controls are selected at the end of follow-up. Over half of the studies included in our review were published after 2020 and addressed adjustments for sampling bias. As noted above, the methods discussed here can target estimands beyond conditional ORs, including marginal effects on other scales. However, in our literature search, most studies reported only conditional ORs.

Each of the statistical methods presented have differing strengths and limitations. Intercept-adjusted logistic regression is simple to use. However, in practice it is rarely used in our identified applications, despite a long history in the literature. Further, it alone cannot be used to derive marginal causal effects because the covariate distribution in the study sample does not represent the covariate distribution in the target population. Simulation studies have shown that this approach performs well in the presence of good model

specification²⁶ but is biased under misspecified models.⁸ These findings follow a larger pattern that outcome-regression-based, causal inference method (like g-computation) is sensitive to outcome model specification because non-linearity and non-additivity must be explicitly addressed. Another outcome-regression based approach, CCW-MLE, is appealing because it is intuitive: sampling weights address outcome and covariate sampling bias by making the weighted case-control sample resemble a cohort study (thus permitting estimation of parameters outside of ORs), and weighted analysis is possible in many software packages. Once weights are applied, it becomes possible to perform additional analyses such as mediation analysis^{40–43}, that are typically not straightforward within the standard case-control framework.

In contrast, propensity score-based methods do not need to specify a model for the outcome and may be less sensitive to model misspecification, but, without modification, do not address outcome sampling bias and limit the scope of available target estimands. Propensity score approaches were the most used approaches identified in our review. We intuit this is not due to an interest in marginal ORs but rather due to their ease of use and an implicit association between “propensity score” and “causal inference,” even though causal inference is made possible via assumptions and study design, rather than particular methods.⁴⁴ Nevertheless, these methods can be biased under propensity model misspecification, and can be sensitive to the positivity assumption because they do not allow interpolation. Simulation studies have also highlighted a potential modification artifact between exposure and estimated propensity scores in moderate sample sizes.²⁹

TMLE incorporates CCW-MLE and propensity scores and shares benefits of both (availability of many target estimands, efficiency of outcome-regression based approaches and misspecification-robustness of propensity score-based approaches). It is also doubly-robust, thus reducing concerns of misspecification. It is conceptually the most difficult method in our review, but the availability of an existing R package⁴⁵ has facilitated its applications. Like CCW-MLE, TMLE allows several parameters such as the risk difference to be estimated, unlike the intercept-logistic adjustment and IPTW (without further modifications). From a public health perspective, additive scale effects like risk differences are more relevant than ORs to inform population health burdens and to assess impacts of potential exposures.^{10,46,47}

We identified one study⁴⁸ with an expressed goal to emulate a hypothetical target trial using case-control data. Their findings indicated that appropriate case-control sampling yields estimates comparable to those derived from cohort studies. Careful consideration in the target trial framework⁴⁹ represents an exciting future direction for case-control studies, which may encourage further exploration of issues like recall bias or selection bias that are more prominent in retrospective studies.

Although the main stated purpose of many of the causal inference methods that have been developed for case-control studies is to adjust for sampling bias, more than a third of the included studies did not mention this specific purpose. In addition, causal assumptions were not discussed in around 44% of the studies. We note that many studies that used the propensity score method made no explicit reference to causal inference. Confounder

control under the exchangeability assumption may have been the primary goal in these studies, where propensity-score-based methods may sometimes excel relative to alternative approaches. If one is interested in a causal OR (if it can be considered causal⁵⁰), then such methods may suffice. All methods we presented assume that the risk/prevalence of the disease in the population (and thus the sampling probabilities, marginally or within strata) is either known or is rare enough to be negligible. However, neither may be the case. Thus, if possible, sensitivity analyses should be performed using a range of plausible prevalence values,¹⁰ especially when the target parameter is a risk difference, which may be particularly sensitive to assumptions about the population risk. Most included studies were matched. Matching in case-control studies can help increase efficiency but does not remove confounding and may even introduce selection bias that needs to be suppressed by appropriate analytical methods.^{19,51,52}

Finally, we identified large gaps in existing methods that accommodate incidence density sampling. Using CCW-TMLE, van der Laan extends some methods for cumulative-incidence-sampled studies to the context of incidence density sampling.⁷ One issue raised is that such studies may be from open cohorts, which present difficulties for potential outcomes-based methods which have been developed within closed populations. A second issue is that, while CCW-TMLE methods can be applied on a time-specific basis (e.g. the time a case is sampled), this works only when many cases arise simultaneously, which may not occur in many sampling schemes (e.g. sampling at exact age of diagnosis). Takeuchi et al consider causal hazard ratios via marginal structural Cox models in a nested case-control study, where time-varying propensity score models (given time-varying confounders) are fit using inverse probability of sampling weights.^{53,54} Such approaches represent a promising avenue for more general causal inference approaches in case-control studies, and it may be beneficial to consider novel study designs that maximize efficiency when time-varying confounders may be of concern.

It is important to clarify that we focused on reviewing existing methods, rather than on a systematic review of all the papers applying these methods. In this sense, we have only considered the Medline database to give to readers an idea of the application of methods in the literature. While we may have missed some relevant applications, it seems unlikely that our search strategy identified a non-representative sample of papers. Also, other methods of causal inference, such as Mendelian randomization have been used in case-control studies, but are not presented here because they rely on different sets of assumptions from the methods that we discussed herein.

In conclusion, this review aims to compile and review existing causal inference methods for case-control studies, fostering a better understanding to promote their application and spur further thinking about methodologic gaps. For this purpose, we provide detailed R code for many of the reported methods (available at <https://github.com/alexpkel1/Case-control-causal-review>). A key factor driving the growth of causal inference methods in cohort studies is the ability to address time-varying confounding for exposures that change over time. Nevertheless, our review identified only three studies^{48,53,55} that have confronted these challenges, highlighting a pressing need for further methodological advancements. As public health challenges become increasingly complex, but no less costly to study, the ongoing

development and application of causal inference methods for case-control studies will be essential for informing effective interventions and making evidence-based policy decisions.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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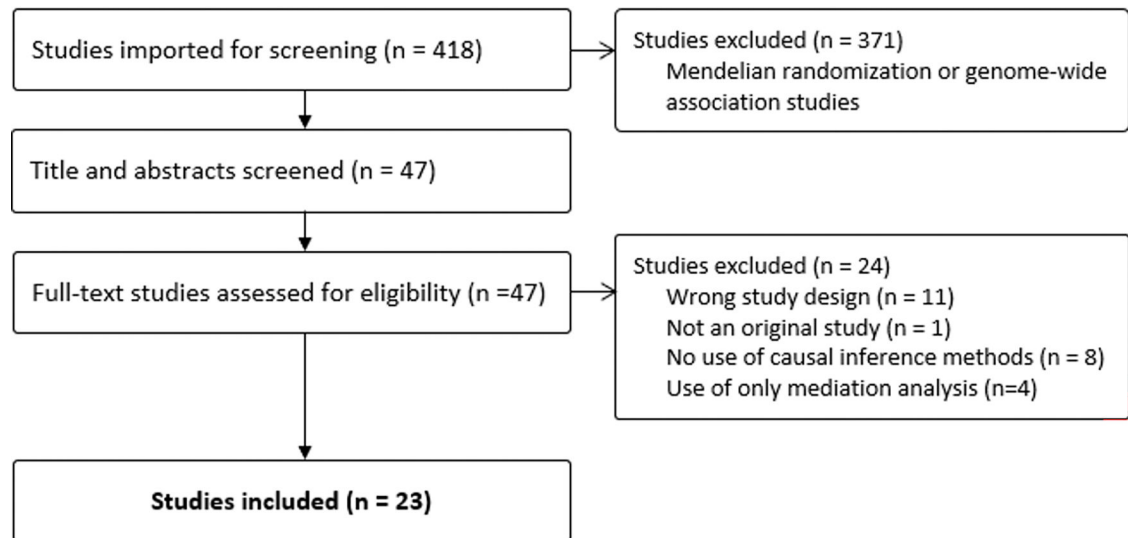


Figure 1.
Preferred reporting items for systematic review and analysis (PRISMA) diagram

Table 1.

Summary of the studies in this review

First author, year	Country	Study design	Types of case-control studies	Recruitment of cases	Parameter estimated	Method used	Adjusted for sampling bias	Discussion of causal assumptions
Backenroth 2016 ⁵⁶	USA ³	Case-control	NA ⁴	NA ⁴	OR ¹	Lasso logistic regression	No	No
Menvielle 2016 ⁵⁷	France	Case-control	Matched	Incident cases	OR ¹	MSM ⁹ ; IPTW ² ; Mediation	Yes	Yes
Pearl 2016 ¹⁰	USA ³	Nested case-control	NA ⁴	Incident cases	RR ⁵ ; RD ⁶	CCW-MLE ⁷ /g-estimation; CCW-TMLE ⁸	Yes	Yes
Bedard 2017 ⁵⁵	France	Nested case-control	Matched	Incident cases	OR ¹	MSM ⁹	No	Yes
Delcoigne 2017 ³⁸	Sweden	Nested case-control	Matched	Incident cases	HR ¹⁰ ; AR ¹¹	IPTW ²	Yes	No
Perisson 2017 ²⁶	Sweden	Case-control	Matched	Incident cases	OR ¹	Intercept-adjusted; CCW-MLE ⁹ /g-estimation; CCW-TMLE ⁸	Yes	Yes
Abdollahpour 2018 ⁵⁹	Iran	Case-control	Matched	Incident cases	OR ¹	IPTW ²	Yes	Yes
Abdollahpour 2018 ⁶⁰	Iran	Case-control	Unmatched	Incident cases	OR ¹	IPTW ² ; MBS ¹²	Yes	Yes
VelasquezGarcia 2018 ⁶¹	Canada	Case-control	Matched	Incident cases	OR ¹	IPTW ²	No	Yes
Zhu 2019 ³¹	USA ³	Case-control	Matched	NA ⁴	OR ¹	Propensity scores	Yes	Yes
Dickerman 2020 ⁴⁸	UK	Nested case-control	Unmatched	Incident cases	OR ¹	Targeted trial	--	No
Figuerola 2020 ⁶²	Costa Rica	Case-control	Matched	Incident cases	OR ¹ ; RR ⁵ ; RD ⁶	CCW-TMLE ⁸ ; targeted learning framework	No	No
Lu 2020 ⁶³	USA ³	Case-cohort	Unmatched	NA ⁴	OR ¹	Propensity scores; IPTW ²	Yes	Yes
Abdollahpour 2021 ⁶⁴	Iran	Case-control	Unmatched	Incident cases	OR ¹ ; RR ⁵ ; RD ⁶ ; PAF ¹³	CCW-TMLE ⁸	Yes	No
Almasi-Hashiani 2021 ⁶⁵	Iran	Case-control	Matched	Incident cases	RR ⁵ ; RD ⁶ ; PAF ¹³	CCW-TMLE ⁸	Yes	No
Basnet 2021 ⁶⁶	Nepal	Case-control	Matched	NA ⁴	OR ¹	IPTW ² ; CCW-MLE ⁹ /g-estimation	No	No

First author, year	Country	Study design	Types of case-control studies	Recruitment of cases	Parameter estimated	Method used	Adjusted for sampling bias	Discussion of causal assumptions
Boska 2021 ⁶⁷	USA ³	Case-control	Matched	Incident cases	OR ¹	Propensity scores	No	No
Akhtar 2022 ⁶⁸	Kuwait	Case-control	Matched	NA ⁴	OR ¹ ; RR ⁵ ; RD ⁶ ; pPF ¹⁴	Propensity scores; MSM; CCW-TMLE ⁸	Yes	Yes
Yu 2022 ⁶⁹	South Korea	Nested case-control	Matched	NA ⁴	OR ¹	Propensity scores	No	No
Elduma 2023 ⁷⁰	Sudan	Case-control	NA ⁴	Incident cases	RR ⁵ ; RD ⁶	CCW-TMLE ⁸	Yes	Yes
Malekifar 2023 ⁷¹	Iran	Case-control	NA ⁴	Incident cases	RR ⁵ ; RD ⁶ ; PAF ¹³	MBS ¹²	Yes	No
Caubert 2024 ⁷²	Canada	Case-control	Matched	Incident cases	OR ¹	IPTW ² ; Mediation	Yes	Yes
Takeuchi 2024 ⁵³	Japan	Nested case-control	Matched	Incident cases	HR ¹⁰	MSM ⁹	Yes	Yes

¹ Odd-ratio

² Inverse probability of treatment weighting

³ United States of America

⁴ Information not available

⁵ Relative risk

⁶ Risk difference

⁷ Case control weighted-maximum likelihood estimation

⁸ Case control weighted-targeted maximum likelihood estimation

⁹ Marginal structural model

¹⁰ Hazard ratio

¹¹ Absolute risk

¹² Model based standardization (surrogate of g-estimation)

¹³ Population attributable fraction

¹⁴ Population preventable fraction